

A global solver for depth-augmented SfM with partial tracks

Notes on the *linear* solver of the sfm box — what was tried, what failed, what works · 2 October 2026

Summary

The input is feature tracks (LightGlue) and a monocular depth map per frame (MoGe), whose depth is trusted only up to a per-frame affine map. Tracks are short, so 50–80 % of the (frame, point) entries are missing, in a banded pattern. The existing method — rank-4 matrix completion alternated with a per-frame affine fit, followed by factorization and metric upgrade — works on complete tracks but breaks down on such data. Splitting the sequence into small blocks and stitching them does not fix it either. The proposed solver uses the fact that, **once the rotations are known, the model is linear in all other unknowns**. Rotations come from 2-frame solves (which never have missing entries), one linear least-squares solve gives translations and points, and a joint Levenberg-Marquardt refinement over all unknowns on a scale-invariant residual finishes the job. Missing entries simply contribute no equations: there is no completion and no stitching. On a synthetic sequence with 77 % missing entries it recovers the cameras exactly without noise and to 0.8° / 0.8 % of the path length with noise. On a real kitchen video it gives 2–6 px reprojection error and a consistent trajectory for every track-length threshold, where the completion baseline fails (65–2467 px, camera flips).

1 Model

For every observed entry (frame f , point p) $\in \Omega$, with normalized ray $\mathbf{w}_{fp} = [x, y, 1]^T$ and monocular depth λ_{fp} :

$$(d_f \lambda_{fp} + o_f) \mathbf{w}_{fp} = R_f X_p + t_f \quad (1)$$

The unknowns are the camera poses (R_f, t_f), the points X_p , and the per-frame depth affine (d_f, o_f). The only true ambiguity is a global similarity (rotation, translation, scale). The offsets are *not* a gauge: they are determined through parallax only, so they become weakly observable when the baseline is small. This matters throughout.

2 Why the existing approaches fail

2.1 Global completion (“ping-pong”)

Rank-4 ALS completion of $(\lambda + o) \cdot \mathbf{w}$ alternated with a per-frame affine fit, RANSAC column rejection, scale correction and factorization. Its start is an SVD of the column-mean fill. With banded tracks this start is far from the solution: the camera estimate degrades sharply from about 40 % missing (rotation jumps of 68° on the real video), and at 77 % missing the synthetic mean rotation error is 64–82°. RANSAC at iteration 10 also rejects gappy columns precisely because their fill is not yet converged. Additionally, the anchor $o_0 = 0$ is not a gauge and biases the result ($\approx 0.6^\circ$ on noiseless synthetic data).

2.2 Overlapping blocks + stitching

The sequence is split into blocks of k frames, each reconstructed independently, and consecutive blocks are aligned by a 4×4 transform fitted on shared points. Three problems:

- **The fitted transform is affine, not projective.** $\text{lstsq}(S_b^T, S_a^T)$ with homogeneous coordinate 1 on both sides solves the fourth row exactly by $[0, 0, 0, 1]$ (measured: 10^{-16}). Each block is already metric, so the correct alignment is a 7-dof similarity; the 5 extra dof (shear, anisotropic scale) enter the cameras through $P \cdot H^{-1}$, which are then no longer scaled rotations.

- **Per-block offset anchors make the blocks mutually inconsistent.** A frame shared by blocks a and b gets two different affine maps (block b anchors its own first offset to 0), so the same point in that camera is reconstructed as

$$X^a = X^b \frac{d^a \lambda + o^a}{d^b \lambda + o^b} \quad (2)$$

— a stretch along each ray that depends on λ , representable by no similarity, affine or projective map unless $o^a/d^a = o^b/d^b$. This is the 6 % residual path error of noiseless 2-frame stitching even with a similarity transform.

- **Why only 2-frame blocks work.** A 2-frame block has no missing entries, so the completion never runs. With 3–6 frames per block, gappy columns appear and the completion fails sporadically: 6 of 18 three-frame blocks had 52–86° rotation error while the rest were $\leq 0.16^\circ$. Since blocks are chained, one bad block ruins every later frame.

3 The key observation: linear given the rotations

With the R_f fixed, (1) contains no product of two unknowns:

$$d_f(\lambda_{fp} \mathbf{w}_{fp}) + o_f \mathbf{w}_{fp} - t_f - R_f X_p = 0 \quad (3)$$

Each observation contributes three linear equations whose coefficients are data. Stacking them gives a sparse homogeneous system $A u = 0$ in $u = (d, o, t, X)$; the gauge $d_0 = 1, t_0 = 0, R_0 = I$ turns it into an ordinary least-squares problem. A point seen in a single frame adds 3 equations and 3 unknowns (no information), so ≥ 2 observations per point are required, and frames must be connected by shared points. On the synthetic test (20 frames, 1009 points, 77 % missing) this is $\approx 13\,900$ equations for 3 123 unknowns, and with the true rotations it recovers d and o to 10^{-11} . The rotations are exactly what 2-frame solves get right (0.02–0.03° error per pair on noiseless synthetic data).

4 What did not work, and why

Getting from the observation above to a solver that works on real video took four corrections. Each one exposed a property of the problem.

- **Alternating linear solve / per-frame Procrustes stalls.** Each half-step is exact, but together they crawl along the direction coupling the offsets and the rotations: on noiseless data o_0 moved by 0.001 in 2 900 iterations while the residual was already 10^{-7} (the “flatlining” of alternation, cf. Buchanan & Fitzgibbon 2005). A joint Gauss-Newton / LM step with rotation increments $R \leftarrow \exp([\omega]_{\times})R$ converges in ≈ 15 iterations (0.4–0.6 s) to 10^{-15} .
- **The algebraic residual is not scale-invariant.** Minimizing (3) weighted by a fixed $1/\lambda_{fp}$ rewards shrinking the scene: on the real video every frame except the gauge frame converged to $d_f \approx 0.15$, leaving frame 0 with 104 px error against ≈ 6 px elsewhere. A mean-depth gauge only moved the degeneracy to the opposite extreme (frame 0's depth flattened to a constant).
- **The offsets need a weak prior.** Even with a scale-invariant residual, at small baseline the offsets are nearly unobservable and drift towards a degenerate solution (depths $\rightarrow \infty, o_0 \approx 74$ with a median depth of 1.2 m).
- **Robustness.** Huber IRLS handles outlier tracks. A redescending loss (Cauchy) and a small weight on the depth row both converge to wrong minima on small-baseline video (up to 35–80° with outliers): with little parallax, the depth row is what conditions the problem.

5 The solver

Step 1 — rotations. Each consecutive pair $(f, f+1)$ is reconstructed on its co-visible points with the existing factorization pipeline; only the relative rotation is kept, and the rotations are chained.

Step 2 — linear initialization. Per-frame scales are initialized from the median depth ratio of points shared by consecutive frames, $o = 0$. With R and $Z = d_f \lambda$ fixed the residual below is linear in (t, X) : one exact solve.

Step 3 — joint refinement. Levenberg-Marquardt over (d, o, R, t, X) on the residual

$$r_{fp} = \mathbf{w}_{fp} - \frac{R_f X_p + t_f}{d_f \lambda_{fp} + o_f} \quad (4)$$

— the ray minus the predicted point divided by the *corrected* depth. Its x/y components are $\approx$ image-plane errors, its z component a relative depth error; scaling (d, o, t, X) jointly leaves it unchanged, so $d_0 = 1$ is a pure gauge and o_0 is estimated, not anchored. The cost is

$$\min \sum_{(f,p) \in \Omega} \rho(\|D r_{fp}\|) + \alpha^2 \sum_f n_f \left(\frac{o_f}{\lambda_{\text{med}}} \right)^2, \quad D = \text{diag}(1, 1, \beta) \quad (5)$$

with ρ the Huber loss (IRLS, scale $2.5 \times 1.4826 \times \text{MAD}$), $\beta = 1$ the depth-row weight, n_f the observations in frame f , λ_{med} the median depth and $\alpha = 0.03$. The point blocks of the normal equations are 3×3 and independent, so each step eliminates the points by the Schur complement (as in bundle adjustment) and solves only an $8F \times 8F$ system for the camera unknowns (d, o, t, ω) :

$$(U - WV^{-1}W^\top) \delta c = -g_c + WV^{-1}g_X, \quad \delta X = -V^{-1}(g_X + W^\top \delta c) \quad (6)$$

Implementation: numpy only, 20 frames and 1-3 k tracks take 5-10 s on CPU (mostly the 2-frame solves). It is exposed in the box as solver: "linear".

6 Results

6.1 Synthetic data (ground truth)

Video-like scene: 20 frames, camera translating and rotating $\approx 2^\circ$ per frame, tracks living 3-8 frames (77 % of the entries missing), per-frame depth affine $d \in [0.7, 1.4]$, $o \sim \mathcal{N}(0, 0.4^2)$; noise: 2 % multiplicative on depth, 0.002 on the rays. Errors: mean / last-frame rotation error, and camera-centre error at the last frame after a scale fit, as a percentage of the path length.

method	noiseless: rotation	noiseless : path	noisy: rotation	noisy: path
global completion (baseline)	64° / 107°	97 %	82° / 93°	99 %
2-frame blocks, affine H (original stitching)	0.21° / 0.23°	23 %	1.61° / 2.83°	8.8 %
2-frame blocks, similarity H	1.08° / 1.74°	5.3 %	1.18° / 2.25°	3.5 %
4-frame blocks, affine H	25° / 42°	79 %	27° / 20°	96 %
pairs: chained 2-frame + rigid global completion	—	—	0.70° / 0.72°	26 %
linear (this work)	0.15° / 0.22°*	0.2 %*	0.77° / 1.44°	0.8 %

* with the default offset prior $\alpha = 0.03$ (true offsets here are large, std 0.4); with $\alpha = 0$ the noiseless recovery is exact (10^{-7}°). On the smaller test scene used by the box tests (10 frames, banded missing + 0.003 ray noise) the error is 0.13° against 0.69° for the completion baseline.

6.2 Real video (cozinha.mp4, LightGlue + MoGe, no ground truth)

Without ground truth we report the median reprojection error of the reconstructed points over all observed entries, and the multi-view consistency of the depth correction: the mean spread of a track's corrected back-projections $R_f^\top((d_f \lambda + o_f) \mathbf{w} - t_f)$ across the frames that see it, as a percentage of the median depth.

sequence (20 frames)	missing	completion	pairs	linear
every 2nd frame, tracks ≥ 2 frames	58%	65.8 px $\cdot$ 2.17 %	9.9 px $\cdot$ 0.90 %	2.3 px $\cdot$ 0.40 %
every 8th frame, tracks ≥ 4 frames	52%	200.7 px $\cdot$ 1.09 %	14.2 px $\cdot$ 1.30 %	4.8 px $\cdot$ 0.67 %
every 8th frame, tracks ≥ 2 frames	68%	2467.0 px $\cdot$ 321.45 %	14.5 px $\cdot$ 1.17 %	4.3 px $\cdot$ 0.53 %
every 15th frame, tracks ≥ 4 frames	66%	124.5 px $\cdot$ 2.19 %	34.2 px $\cdot$ 4.16 %	6.2 px $\cdot$ 0.72 %

Cells: median reprojection error $\cdot$ 3D spread. All three solvers run in the same box on the same inputs; completion and pairs keep fewer points (RANSAC / visibility).

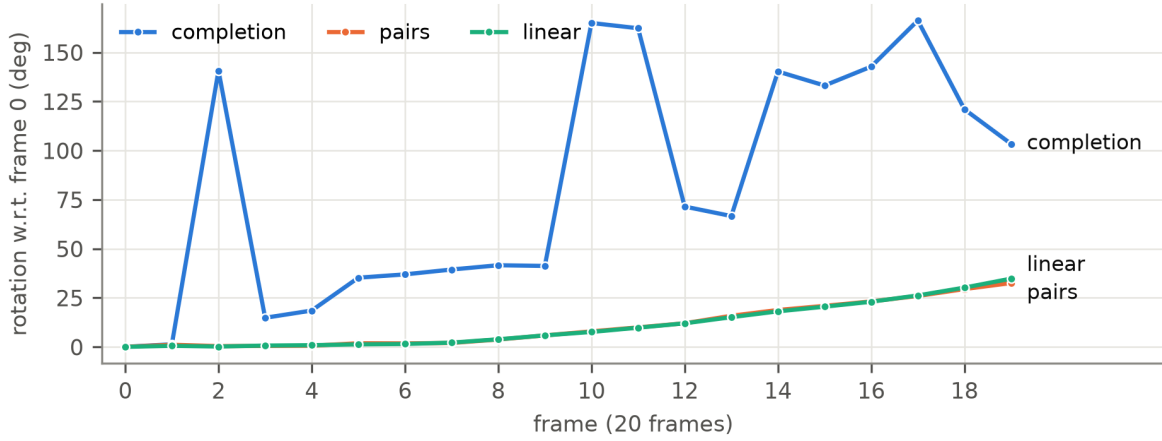


Figure 1. Camera rotation relative to frame 0, every 8th frame, tracks ≥ 2 frames (68 % missing). Completion flips by up to 175° between frames; pairs and linear give the same smooth trajectory ($\approx 2^\circ$ per frame), at 14.5 px and 4.3 px reprojection error respectively.

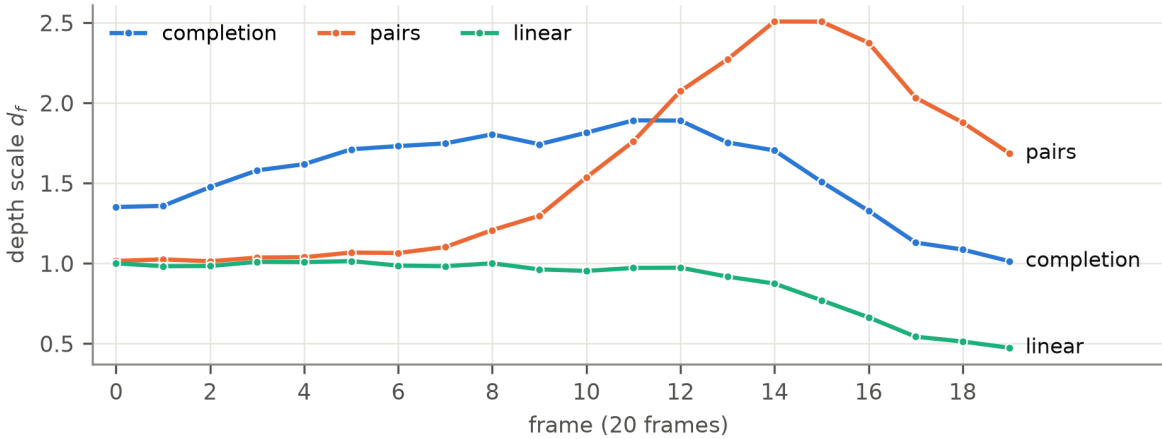


Figure 2. Estimated per-frame depth scale d_f every 15th frame (the camera turns $\approx 90^\circ$ and approaches the scene; the MoGe median depth falls from 1.27 m to 0.52 m). Completion and pairs drift to $d \approx 1.9$ -2.5, inflating the later depth maps; linear stays at ≈ 1 until frame 13 and then falls to 0.47 — see the open questions.

7 Open questions for discussion

- **Observability of the offsets.** They are fixed only by parallax; a prior is unavoidable on small-baseline video and biases the solution when the true offsets are large (0.15° in the noiseless synthetic case). Is there a principled choice of α , e.g. from the conditioning of the (o, X) block, or a reparametrization that separates the observable part?

- **Per-frame scale drift.** On the 90° sequence d_f falls to 0.47 for the closest views. Either MoGe's scale is inconsistent at close range, or the solution drifts. Since MoGe is metric, a weak prior $d_f \rightarrow 1$ would be the analogue of the offset prior — but it would hide genuine scale errors of the depth network.
- **Noise model.** β weighs the depth row against the image rows; monocular depth error is structured (edges, distance-dependent), not i.i.d. Should the depth row be weighted per pixel (e.g. by MoGe's confidence or $1/Z$)?
- **Rotation initialization.** Only consecutive pairs are chained. Adding $(f, f+2)$, $(f, f+3)$ and rotation averaging would reduce drift and remove the single point of failure of a bad pair.
- **Evaluation.** The real-video numbers are self-consistency measures. A sequence with ground truth poses (e.g. ETH3D, already prepared in the trackers repository) is needed to compare the solvers properly.
- **Relation to the literature.** Step 2 is the known-rotation linear problem (cf. Kahl & Hartley; Cui et al.) augmented with per-frame depth affines; step 3 is a bundle adjustment with monocular-depth terms. Is the scale-invariant residual (4) the right measurement model, or should image and depth errors be separated explicitly?

References

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Appendices — findings to return to

Exploratory results, recorded for later discussion. Same data and solver as Section 6.2 (cozinha.mp4, 20 frames, LightGlue + MoGe, *linear* solver); not yet part of the method.

A Dropping the offset: $\mathbf{Z} = \mathbf{d}_f \lambda$

Why the problem becomes much easier. Without o_f , each frame's depth map is an exact 3D point cloud up to one scale, $P_{fp} = \lambda_{fp} \mathbf{w}_{fp}$, and the model reads

$$d_f P_{fp} = R_f X_p + t_f \quad (7)$$

— registration of per-frame point clouds with known correspondences, one similarity per frame (generalized Procrustes analysis). Consequences:

- **Pairs in closed form.** A 3D-3D similarity (Umeyama) from ≥ 3 correspondences; RANSAC with 3-point samples; no factorization, no metric upgrade, no mirrored-camera sign ambiguity.
- **No parallax needed for the scales.** The ratio d_{f+1}/d_f is fixed by any shared points, even under pure rotation. The offsets were the only unknowns that needed parallax.
- **The weakly observable direction disappears.** Every failure mode of Section 4 went through the offsets: the scene-shrinking collapse ($o_0 \rightarrow -\lambda$), the flattening under the mean-depth gauge, the runaway depths without a prior. Without o , the fully linear solve of (d, t, X) given the rotations needs no prior and no special gauge, and alternating it with per-frame Procrustes is the well-behaved GPA iteration.

What it costs on real data. $o = 0$ is imposed with a very strong offset prior ($\alpha = 10^4$); “free” is $\alpha = 0$.

sequence (20 frames)	with o ($\alpha = 0.03$)	$o = 0$	o free ($\alpha = 0$)
every 2nd frame, tracks ≥ 2	2.31 px · 0.9°	2.68 px · 0.8°	4.15 px · $ o = 82$ (runs off)
every 8th frame, tracks ≥ 2	4.34 px · 34.7°	4.62 px · 30.6°	6.66 px · 36.7°
every 8th frame, tracks ≥ 4	4.77 px · 35.3°	4.90 px · 30.8°	7.33 px · 37.2°
every 15th frame, tracks ≥ 4	6.24 px · 87.9°	7.05 px · 86.8°	8.02 px · 86.9°

Cells: median reprojection error · total camera rotation over the sequence. The 3D spread (Section 6.2) is nearly unchanged with $o = 0$ (0.40 → 0.43 %, 0.67 → 0.70 %, 0.72 → 0.79 %).

- **The fit barely needs the offsets** (+0.1 to +0.8 px): MoGe's metric depth has little shift error.
- **The trajectory does change:** about 4° less total rotation at step 8 (30.7° against 35°). Small offsets still matter geometrically; which trajectory is right needs ground truth.
- **The scale drift of Figure 2 is not caused by the offsets:** with $o = 0$, d_f still falls to 0.56 on the 90° sequence.
- **Free offsets are worse than both** — the weak observability of Section 4.

Possible design. (i) Solve with $o = 0$ by the closed-form, robust registration pipeline; (ii) use it as the starting point; (iii) free the offsets with the weak prior as a final refinement. Intermediate model: one offset shared by all frames, far better constrained than one per frame.

B Estimating the focal length

Observability. With unknown focal f (principal point at the image centre) the ray is

$$\mathbf{w}_{fp}(f) = \left[\frac{u_{fp} - c_x}{f}, \frac{v_{fp} - c_y}{f}, 1 \right]^T \quad (8)$$

A wrong f stretches every frame's back-projected cloud laterally — x and y scale by f_{true}/f — but not in depth. When the camera rotates, the stretched axes of different frames no longer align

and the frames become inconsistent: f is observable. Under pure translation the ambiguity is exact (stretching the scene by $\text{diag}(k, k, 1)$ and the translations alike explains the data); a rotation about the optical axis only is degenerate as well.

Profile on the real video. The linear solver was re-run with MoGe's focal multiplied by k ; the median reprojection error is measured in pixels at that focal.

$k \times \text{MoGe focal}$	0.7	0.85	1	1.15	1.3	1.5
every 2nd frame (0.9° rotation)	2.27	2.28	2.32	2.35	2.40	2.45
every 8th frame (35°)	7.36	5.93	4.77	4.38	4.84	5.89
every 15th frame (88°)	9.72	7.96	6.24	5.27	5.80	7.61

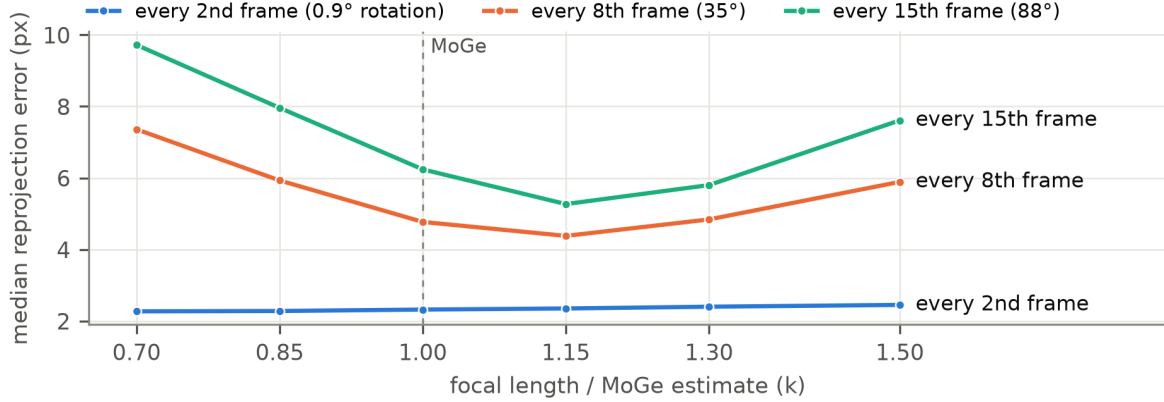


Figure 3. Focal profile. Flat without rotation (every 2nd frame: f unobservable); both rotating sequences have their minimum at the same place, $\approx 1.15 \times \text{MoGe's focal}$ (normalized $f_x \approx 0.88 \rightarrow \approx 1.0$).

- **Reading.** Two independent sequences agree: MoGe probably underestimates this camera's focal length by $\approx 15\%$. Caveats: coarse grid; MoGe predicted its depth assuming its own focal, so its depth maps may not be fully consistent with another f .
- **Adding it to the solver.** One global unknown f (one per frame if the camera zooms) in the LM of Section 5: a single column coupling all frames, i.e. an $(8F + 1)$ bordered Schur system at no real cost. The image rows of the residual must then be measured in **pixels** (multiply the x/y rows by f): in normalized units the residual shrinks just by increasing f — the same scale trap as the offsets. Keep the principal point at the centre (weakly observable), and add a weak prior towards MoGe's focal: f (a lateral stretch) and o (a depth-dependent radial stretch) are distinct but correlated at narrow fields of view.
- **Related work.** Classical self-calibration from rotation (Kruppa equations; Pollefeys et al.); single-image recovery of focal and depth shift from a point map (LeReS, Yin et al.; MoGe, Wang et al.). The multi-view constraint here should be considerably stronger than the single-image one.